AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed Systems

Final Examination
Fall Semester 2025/2026

Date: 14/Jan./2026

Number of Exam Pages: (3) (including this cover sheet)	Time Allowed: (2) Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (10 Marks, 2 marks for each)

1. Which of the following best describes Cassandra's architecture?
   a) Master–slave architecture
   b) Peer-to-peer architecture
   c) Centralized shared memory architecture
   d) Single-leader replicated architecture

2. In the CAP theorem, Cassandra belongs to which category?
   a) CA – Consistency & Availability
   b) CP – Consistency & Partition Tolerance
   c) AP – Availability & Partition Tolerance
   d) AC – Atomicity & Consistency

3. Which AWS service provides virtual machines in the cloud?
   a) Amazon RDS
   b) Amazon EC2
   c) Amazon S3
   d) Amazon VPC

4. In Cassandra, which key determines how data is distributed across nodes?
   a) Partition key
   b) Clustering key
   c) Compound key
   d) Primary key only

5. Which Docker command is used to pull the latest Cassandra image?
   a) docker get cassandra
   b) docker download cassandra:latest
   c) docker pull cassandra:latest
   d) docker install Cassandra

Part II: Short Answer Questions (10 marks, 3+4+3)

1) Explain the difference between SimpleStrategy and NetworkTopologyStrategy in Cassandra replication.

2) List three advantages of using AWS cloud computing compared to traditional on-premises infrastructure.

3) Define Partition key and Clustering key in Cassandra and state their difference.

Part III: Problem-Solving Question (30 marks)

Scenario:
You are asked to deploy a 2-node Cassandra cluster using Docker for a real-time analytics system. The system must:

- Use NetworkTopologyStrategy with RF=2

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- Store data in a keyspace named analytics
- Include a table events with partition key sensor_id and clustering key timestamp (descending order)
- Be secured with PasswordAuthenticator and a role called data_engineer with INSERT and SELECT permissions
- Be tested using nodetool status

Questions:

a) Write the Docker commands to start two Cassandra nodes, where the second node joins the first node using seeds. (4 marks)

b) Write the CQL command to create the keyspace analytics with RF=2 using NetworkTopologyStrategy. (4 marks)

c) Create the table events using the required partition and clustering keys. (4 marks)

d) Enable password authentication in cassandra.yaml. (3 marks)

e) Create the role data_engineer with password and grant permissions. (4 marks)

f) Show the command to verify cluster health. (2 marks)

g) Explain any two reasons why Cassandra has no single point of failure. (2 marks)

h) During a write request, what happens if one replica node is down in Cassandra using the AP model? (2 marks)

i) Write the CQL command to increase the replication factor from 2 to 3 for keyspace analytics without losing data. (2 marks)

j) Why are joins avoided in Cassandra data modeling? Give one reason only. (1 mark)

k) Analyze how AWS achieves the high availability through its global infrastructure and pricing model. (2 marks)

End of questions

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